

MICROCANONICAL PREPARATION AND EXPONENTIAL-SCALE ENSEMBLE TRANSFER FOR DETERMINISTIC HARD SPHERES

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ABSTRACT. We derive the source-dependent microcanonical denominator directly from the positive interacting exact-particle-number hard-sphere integral. No independence of connected components and no probabilistic interpretation of signed polymer activities are used. Sequential physical-cell exposure and velocity integration by parts give one integrable characteristic-function majorant over all frequencies, a uniformly positive covariance, and a geometric mixed-shell local coefficient theorem. The proof order is strictly grand-canonical B2, then B1 conditioning, then the microcanonical B2 large-deviation theorem.

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1. ROUND-SEVENTEEN POSITIVE CLOSURE: DIRECT CANONICAL COEFFICIENTS AND GEOMETRIC SHELL ASYMPTOTICS

This section proves the exact-particle-number coefficient theorem directly from the positive canonical phase-space integral. Connected trajectory polymers enter only through analytic error bounds for the grand-canonical pressure; they are never interpreted as independent random components.

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The proof separates the logically prior grand-canonical B2 input from the conditioning theorem proved here.

1.1. Exact-number positive law and the noncircular input. Fix a short physical horizon $T \leq T_*$ and a bounded real path/contact source H in the grand-canonical source chart constructed in the grand-canonical part of Paper B2. For $N \geq 1$ put

$$Z_{\varepsilon,N}(H, \lambda) = \frac{1}{N!} \int_{\mathcal{D}_{\varepsilon,N}} \exp \left\{ -\beta E_N + \mu_\varepsilon H_N + \lambda \cdot \sum_{i=1}^N C(z_i) \right\} dZ_N,$$

where $C = (v, |v|^2/2, \chi_1, \dots, \chi_m)$ is the continuous constraint vector after particle number has been fixed. For real (H, λ) this is a strictly positive finite integral and therefore defines the probability $\mathbf{P}_{\varepsilon,N}^{H,\lambda}$.

The only B2 input used before conditioning is the following: on a fixed complex source/activity polydisc, the logarithm of the grand-canonical partition function is normally convergent, its source derivatives through order four are bounded, and its connected remainder after the singleton term is $O(\varepsilon)$ in the low-density norm. No microcanonical conclusion of B2 is used in this section.

Lemma 1.1 (Direct canonical saddle). *Let a range over a compact subset of the interior of the one-particle constraint mean image. For all sufficiently small ε , every real H in a fixed compact source ball has a unique real multiplier $\lambda_{\varepsilon,N,H,a}$ satisfying*

$$\frac{1}{N} \mathbf{E}_{\varepsilon,N}^{H,\lambda} \sum_{i=1}^N C(z_i) = a.$$

Moreover

$$cI \leq \frac{1}{N} \text{Cov}_{\varepsilon,N}^{H,\lambda} \left(\sum_i C(z_i) \right) \leq cI$$

uniformly at the saddle, and the saddle depends analytically on the source.

Proof. For the independent one-particle reference measure, the Hessian of the log-Laplace transform is uniformly positive on compact subsets of the interior mean image: if $b \neq 0$, then $b \cdot C$ is nonconstant on one of a finite family of velocity-position boxes, and its variance there is bounded below. Write the exact canonical integral by integrating particles in a fixed order. At each step the forbidden spatial volume is at most $CN\varepsilon^3 = O(\varepsilon)$, uniformly in the previously exposed configuration. Differentiating the positive integral under the sign of integration and comparing conditional one-particle kernels therefore gives

$$\left\| \frac{1}{N} D_\lambda^2 \log Z_{\varepsilon,N} - D^2 q_1(\lambda) \right\| \leq C\varepsilon$$

on the compact chart. The path/contact source changes the conditional kernel by a uniformly bounded Radon–Nikodym factor; the grand-canonical

connected bound controls its derivative and preserves the same $O(\varepsilon)$ comparison. Strict convexity and the inverse-function theorem give the unique saddle and its analytic source dependence. $\square$

1.2. A direct Fourier decomposition of the positive canonical integral. Let

$$S_N = \sum_{i=1}^N C(z_i), \quad \Phi_{\varepsilon,N,H}(u) = \mathbf{E}_{\varepsilon,N}^{H,\lambda_{\varepsilon,N,H,a}} e^{iu \cdot (S_N - Na)}.$$

Choose finitely many compact coordinate boxes $U_1, \dots, U_J$ away from hard-core boundaries such that, for every unit vector e , one box contains a coordinate direction ξ with $|\partial_\xi(e \cdot C)| \geq c_*$. Partition configuration space according to the lexicographically first accepted particles lying in those boxes. This is a measurable partition of the genuine positive canonical law; it is not a polymer decomposition.

Lemma 1.2 (Canonical cell matching). *There are $c_0, c_1 > 0$ and a family of disjoint mesoscopic spatial cells such that, under every saddle law above, with probability at least $1 - e^{-c_0 N}$ at least $c_1 N$ cells contain one selected particle whose velocity ranges over one of the boxes U_j . Conditional on the exterior and on all selected cell labels, the joint density of the selected velocities is bounded above and below, together with four derivatives, by tensor-product one-particle densities times a factor R satisfying*

$$\|\log R\|_{C^4} \leq C\varepsilon n_{\text{msel}}.$$

On the exceptional event there is still one d -dimensional coarea chart, and its conditional density has $W^{d+3,1}$ norm bounded by C .

Proof. Take cells of side $L\varepsilon$ with L fixed and large. The exclusion neighborhood generated by previously exposed particles occupies a uniformly small fraction of a fresh cell. The exact canonical density is positive and, on a compact real source chart, its conditional activity in an accepted box is bounded above and below. Sequential exposure therefore stochastically dominates a Bernoulli process with fixed success probability; the bounded-difference martingale inequality gives the exponential lower-tail estimate.

Given the selected positions and the exterior, hard-core exclusion is velocity independent. The remaining dependence among selected velocities comes from the bounded dynamic source. Its logarithmic mixed derivatives are sums of connected trajectory contributions. The grand-canonical B2 estimate, used only as an analytic absolute bound, gives the displayed C^4 estimate for R . Finally, the affine-span hypothesis for C provides a finite cover of configuration space by full-rank coarea charts; disintegrating the positive canonical integral over the first available chart proves the exceptional statement. No independence of connected collision components is asserted. $\square$

Theorem 1.3 (One integrable canonical Fourier majorant). *There are $0 < \delta < R < \infty$, $c > 0$, and $s > d + 4$ such that, uniformly in the compact source/target chart,*

$$|\Phi_{\varepsilon,N,H}(u)| \leq \begin{cases} \exp(-cN|u|^2), & |u| \leq \delta, \\ \exp(-cN), & \delta \leq |u| \leq R, \\ Ce^{-cN}(1+|u|)^{-s} + C(1+|u|)^{-s-cN}, & |u| > R. \end{cases}$$

The estimate holds after up to three source derivatives, with a fixed polynomial loss absorbed by choosing s larger. In particular

$$\int_{|u|>\delta} |\Phi_{\varepsilon,N,H}(u)| du = o(N^{-d/2}).$$

Proof. The central estimate follows from Lemma 1.1, Taylor's formula, and the fourth-cumulant bound from the direct positive integral. On the compact annulus, every unit direction is detected in one of the boxes U_j . Conditional integration over one selected velocity gives a characteristic factor whose modulus is at most $1 - c$; applying this successively to cN selected cells, while absorbing the $C\varepsilon N$ interaction factor into the exponent, gives e^{-cN} .

For $|u| > R$, choose in every selected block a coordinate with phase derivative at least $c|u|$. Repeated integration by parts in that physical velocity coordinate is legitimate because the boxes are interior and the partition is chosen with a smooth subordinate cutoff. Boundary terms vanish; derivatives fall on the one-particle density and on R . On the typical event, applying $s + cN$ integrations distributed among the selected particles gives $C(1 + |u|)^{-s-cN}$. On the exceptional event, its probability is e^{-cN} and the fallback coarea chart supplies the fixed integrable factor $(1 + |u|)^{-s}$. These are the two displayed terms; there is no frequency-independent exceptional contribution. Bounded source insertions consume only finitely many integrations. $\square$

1.3. Geometric shell theorem. Let $D_\varepsilon \subset \mathbb{R}^d$ be a bounded domain with uniformly C^2 boundary, uniformly bounded Gaussian perimeter, and inradius ρ_ε satisfying $N^{1/2}\rho_\varepsilon \rightarrow \infty$. This geometric definition does not contain a desired Fourier-error conclusion.

Theorem 1.4 (Exact-number local coefficient). *Uniformly in the compact source/target chart,*

$$\mathbf{P}_{\varepsilon,N}^{H,\lambda_\varepsilon,N,H,a} \{N^{-1}S_N \in a + D_\varepsilon\} = \int_{a+D_\varepsilon} \frac{N^{d/2} \exp[-\frac{N}{2}(y-a)^T \Sigma^{-1}(y-a)]}{(2\pi)^{d/2} \sqrt{\det \Sigma}} dy (1+o(1)),$$

where $\Sigma = N^{-1} \text{Cov}(S_N)$. The error is uniform after three source derivatives. Point boxes, mesoscopic windows, and expanding central boxes follow by the same formula in their natural regimes.

Proof. Fourier inversion is split at δ and R . In the central ball the uniform fourth-order cumulant expansion, after $u = N^{-1/2}z$, converges to the Gaussian integrand and has an integrable Gaussian envelope. The compact and

large-frequency integrals are $o(N^{-d/2})$ by Theorem 1.3. Approximate the indicator of D_ε from above and below by convolution at a scale $N^{-1/2} \ll h_\varepsilon \ll \rho_\varepsilon$. The boundary error is its Gaussian perimeter times $h_\varepsilon/h_{o_\varepsilon} = o(1)$. This proves the statement from geometric hypotheses rather than assuming the needed error. $\square$

1.4. Source-dependent microcanonical transfer.

Theorem 1.5 (Noncircular positive shell conditioning). *Let $Q^N(H, \lambda)$ be the exact-number limiting pressure obtained from the grand-canonical B2-GC pressure by coefficient extraction. For every regular constraint target a ,*

$$Q^{\text{mc},a}(H) = \inf_{\lambda} \{Q^N(H, \lambda) - \lambda \cdot a\} - \inf_{\lambda} \{Q^N(0, \lambda) - \lambda \cdot a\}.$$

The minimizer is reoptimized for every path/contact source H , and

$$D^2 Q^{\text{mc},a} = Q_{HH} - Q_{H\lambda} Q_{\lambda\lambda}^{-1} Q_{\lambda H}.$$

The proof order is strictly

B2-GC analytic pressure $\longrightarrow$ B1 coefficient and shell theorem $\longrightarrow$ B2-MC LDP.

Proof. Tilt the positive exact- N measure by the finite-volume saddle from Lemma 1.1. The shell is centered under the tilted law, and Theorem 1.4 gives a subexponential, uniformly positive Gaussian-scale denominator. Dividing numerator and denominator and passing to the pressure limit yields the two saddle values. Differentiating the exact saddle equation gives the Schur complement. Only the grand-canonical B2 input was used before this point; B2's microcanonical result is therefore downstream rather than circular. $\square$

Theorem 1.6 (Round-seventeen B1 closure). *The exact-number hard-sphere law admits a source-dependent saddle, uniformly nondegenerate constraint covariance, one global integrable Fourier majorant, a geometric mixed-shell local coefficient theorem, and noncircular microcanonical transfer. No signed polymer activity is treated as a probability and no independent connected-component representation is used.*

Proof. Combine Lemmas 1.1 and 1.2 with Theorems 1.3, 1.4, and 1.5. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states a positive theorem reconstruction in the model-specific regular regime declared in its controlling theorems. Cross-paper inputs follow the Round-Seventeen dependency ledger. Repository source identity, proof-structure, counterexample, and build checks are internal reproducibility gates and do not replace independent external mathematical review.

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